

ECON 212 — Principles of Microeconomics

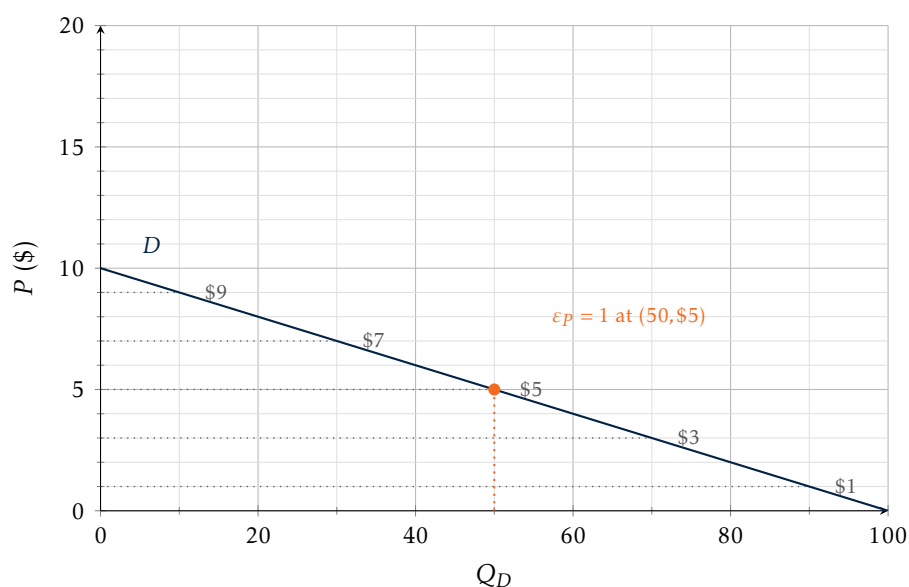
Handout: Elasticity by the Numbers

Fall 2026

Part I. Price Elasticity of Demand

Problem 1. A demand curve, three ranges

Let $Q_D = 100 - 10P$. The three ranges are \$1–\$3, \$3–\$7, and \$7–\$9.



(a) Fill in the demand schedule table.

P	\$1	\$3	\$5	\$7	\$9
Q_D	90	70	50	30	10

(b) Compute ϵ_P by the midpoint method over each of the three ranges.

Answer. \$1–\$3: $\% \Delta Q = -20/80 = -25\%$, $\% \Delta P = 2/2 = 100\%$, so $\epsilon_P = \frac{1}{4}$.
 \$3–\$7: $\% \Delta Q = -40/50 = -80\%$, $\% \Delta P = 4/5 = 80\%$, so $\epsilon_P = 1$.
 \$7–\$9: $\% \Delta Q = -20/20 = -100\%$, $\% \Delta P = 2/8 = 25\%$, so $\epsilon_P = 4$.

(c) Classify each range as elastic, unit elastic, or inelastic.

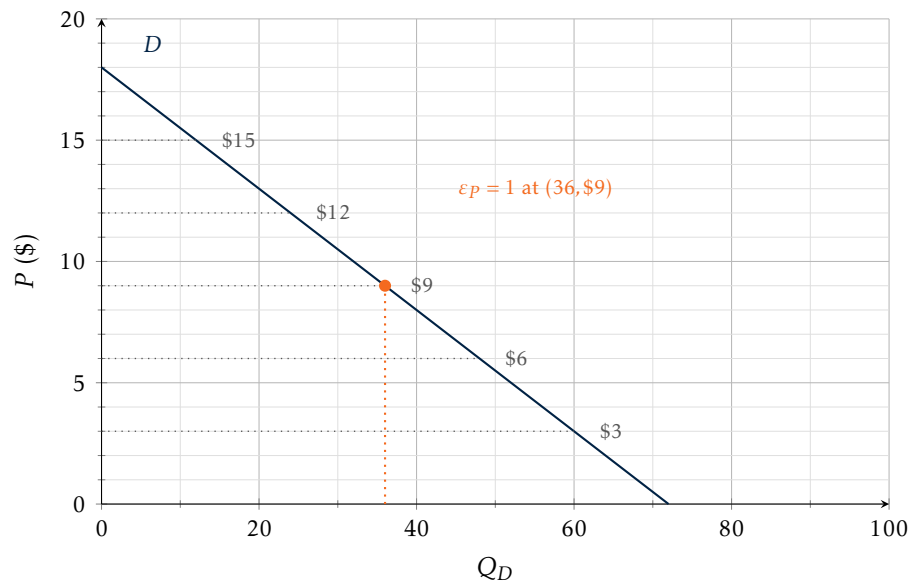
Range	\$1–\$3	\$3–\$7	\$7–\$9
Classification	inelastic	unit elastic	elastic

(d) Mark on the figure the point where demand is unit elastic, and give the (Q, P) pair.

Answer. $(Q, P) = (50, \$5)$ — the midpoint of the \$3–\$7 range, and the midpoint of the curve.

Problem 2. A different curve, three ranges

Let $Q_D = 72 - 4P$. The three ranges are \$3–\$6, \$6–\$12, and \$12–\$15.



(a) Fill in the demand schedule table.

P	\$3	\$6	\$9	\$12	\$15
Q_D	60	48	36	24	12

(b) Compute ϵ_P by the midpoint method over each of the three ranges.

Answer. \$3–\$6: $\% \Delta Q = -12/54 = -22.2\%$, $\% \Delta P = 3/4.5 = 66.7\%$, so $\epsilon_P = \frac{1}{3}$.
 \$6–\$12: $\% \Delta Q = -24/36 = -66.7\%$, $\% \Delta P = 6/9 = 66.7\%$, so $\epsilon_P = 1$.
 \$12–\$15: $\% \Delta Q = -12/18 = -66.7\%$, $\% \Delta P = 3/13.5 = 22.2\%$, so $\epsilon_P = 3$.

(c) Classify each range as elastic, unit elastic, or inelastic.

Range	\$3–\$6	\$6–\$12	\$12–\$15
Classification	inelastic	unit elastic	elastic

(d) Mark on the figure the point where demand is unit elastic, and give the (Q, P) pair.

Answer. $(Q, P) = (36, \$9)$ — the midpoint of the \$6–\$12 range, and the midpoint of the curve.

Part II. The Other Two Elasticities

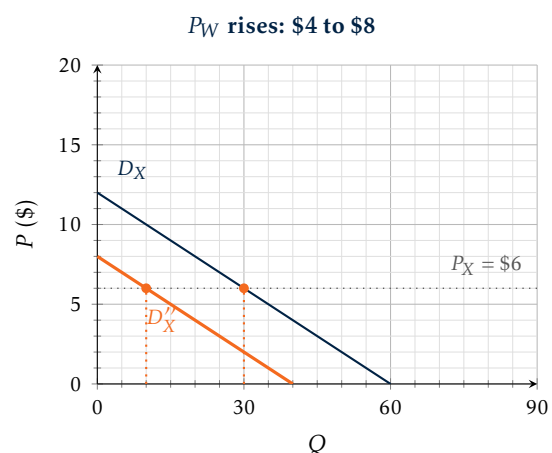
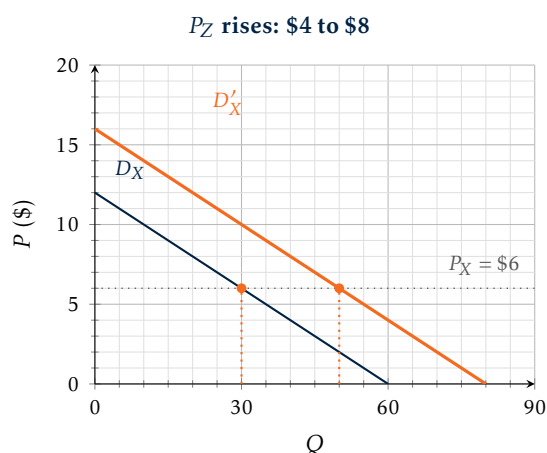
Report ϵ_{CP} and ϵ_I with their sign.

Problem 3. Cross-Price Elasticity

Good X has demand

$$Q_X = 60 - 5P_X + 5P_Z - 5P_W,$$

where Z and W are two other goods. Start at $P_X = \$6$, $P_Z = \$4$, and $P_W = \$4$.



- (a) Compute the starting Q_X at the given prices.

Answer. $Q_X = 60 - 30 + 20 - 20 = 30$.

- (b) P_Z rises to \$8, holding everything else fixed. Compute the new Q_X and ϵ_{CP} by the midpoint method.

Answer. $Q_X = 60 - 30 + 40 - 20 = 50$. $\% \Delta Q = 20/40 = 50\%$; $\% \Delta P_Z = 4/6 = 66.7\%$; $\epsilon_{CP} = +0.75$.

- (c) Instead, P_W rises to \$8, holding everything else fixed. Compute the new Q_X and ϵ_{CP} by the midpoint method.

Answer. $Q_X = 60 - 30 + 20 - 40 = 10$. $\% \Delta Q = -20/20 = -100\%$; $\% \Delta P_W = 4/6 = 66.7\%$; $\epsilon_{CP} = -1.5$.

- (d) Classify Z and W as substitutes or complements.

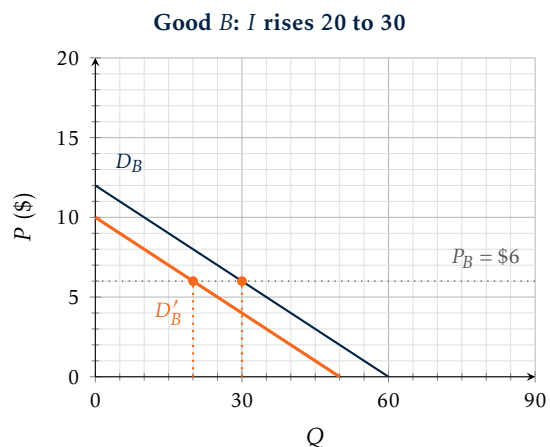
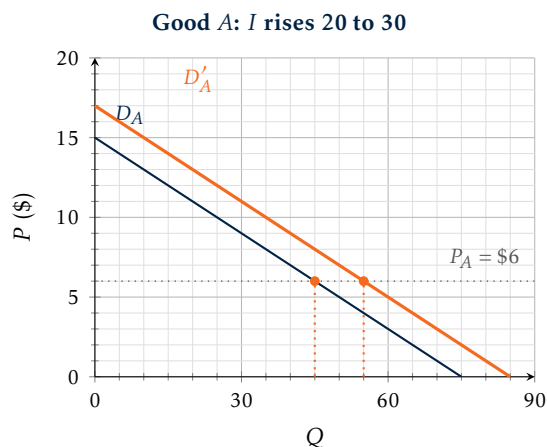
Good	Z	W
Substitute or complement	substitute	complement

Problem 4. Income Elasticity

Goods A and B have demands

$$Q_A = 55 + I - 5P_A, \quad Q_B = 80 - I - 5P_B,$$

where I is income in thousands of dollars. Both prices sit at \$6 and stay there. Start at $I = 20$.



- (a) Compute Q_A and Q_B at the starting income.

Answer. $Q_A = 55 + 20 - 30 = 45$; $Q_B = 80 - 20 - 30 = 30$.

- (b) Income rises to $I = 30$. Compute the new Q_A and Q_B .

Answer. $Q_A = 55 + 30 - 30 = 55$; $Q_B = 80 - 30 - 30 = 20$.

- (c) Compute ε_I for each good by the midpoint method.

Answer. $\% \Delta I = 10/25 = 40\%$ for both.

A : $\% \Delta Q_A = 10/50 = 20\%$, so $\varepsilon_I = +0.5$.

B : $\% \Delta Q_B = -10/25 = -40\%$, so $\varepsilon_I = -1$.

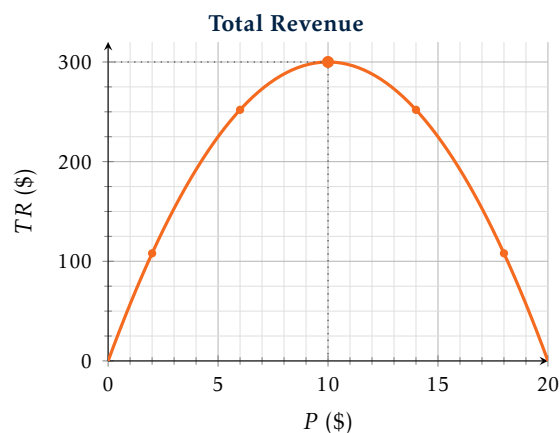
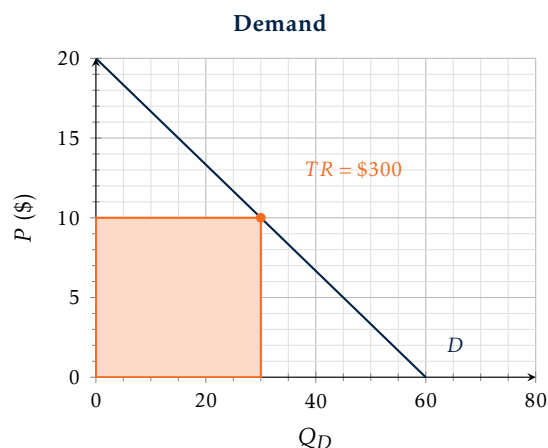
- (d) Classify each good: inferior, normal necessity, or normal luxury.

Good	A	B
Classification	normal necessity	inferior

Part III. Revenue Maximization

Problem 5. Revenue on a steep curve

Let $Q_D = 60 - 3P$. The five prices are \$2, \$6, \$10, \$14, and \$18.



(a) Fill in the table: Q_D and total revenue $TR = P \times Q$ at each price.

P	\$2	\$6	\$10	\$14	\$18
Q_D	54	42	30	18	6
TR	\$108	\$252	\$300	\$252	\$108

(b) At which price is TR largest? What is TR there?

Answer. $P = \$10$, with $TR = \$300$.

(c) Compute ε_P by the midpoint method over the two ranges straddling that price.

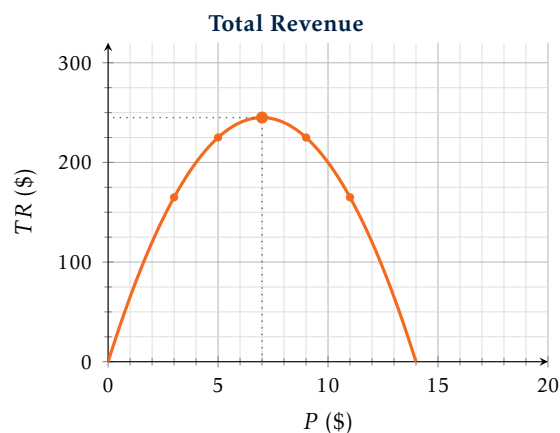
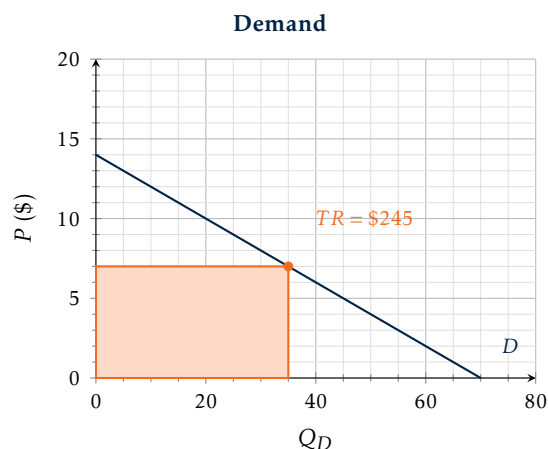
Answer. \$6–\$10: $\% \Delta Q = -12/36 = -33.3\%$, $\% \Delta P = 4/8 = 50\%$, so $\varepsilon_P = \frac{2}{3}$.
 \$10–\$14: $\% \Delta Q = -12/24 = -50\%$, $\% \Delta P = 4/12 = 33.3\%$, so $\varepsilon_P = \frac{3}{2}$.

(d) The firm currently charges \$16. To raise revenue, should it raise or lower its price? Same question at \$4.

Answer. At \$16, lower it. At \$4, raise it. Both roads lead to \$10.

Problem 6. Revenue on a flatter curve

Let $Q_D = 70 - 5P$. The five prices are \$3, \$5, \$7, \$9, and \$11.



- (a) Fill in the table: Q_D and total revenue $TR = P \times Q$ at each price.

P	\$3	\$5	\$7	\$9	\$11
Q_D	55	45	35	25	15
TR	\$165	\$225	\$245	\$225	\$165

- (b) At which price is TR largest? What is TR there?

Answer. $P = \$7$, with $TR = \$245$.

- (c) Compute ε_P by the midpoint method over the two ranges straddling that price.

Answer. \$5–\$7: $\% \Delta Q = -10/40 = -25\%$, $\% \Delta P = 2/6 = 33.3\%$, so $\varepsilon_P = \frac{3}{4}$.
 \$7–\$9: $\% \Delta Q = -10/30 = -33.3\%$, $\% \Delta P = 2/8 = 25\%$, so $\varepsilon_P = \frac{4}{3}$.

- (d) The firm currently charges \$12. To raise revenue, should it raise or lower its price? Same question at \$2.

Answer. At \$12, lower it. At \$2, raise it. Both roads lead to \$7.